SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
JAWAHARLAL NEHRU UNIVERSITY
New Delhi -110067

MCA III SEMESTER

Internal Assessment II (Monsoon) CS 425 Blockchain Technology

Date: 23-09-2025 & Time: 9:00 AM to 11:00 AM **Max Marks: 50**

Question No. 1 to 5 Carry ONE marks each (Answer ALL) **(1x5=5)**

1. Blockchain is a ______________________, ______________________ ledger that is ______________- secure, ______________________-only, ______________________ (extremely hard to change), and updateable only via ______________________ or agreement among peer.
2. Bitcoin ______________ is a computer protocol intended to digitally facilitate, verify, or enforce the negotiation or performance of a contract
3. The Sehilif ______________ is the first and foundational block in a blockchain, hardcoded into its protocol as the starting point for all subsequent blocks and transactions.
4. One wei is equal to Rs. ______________________
5. A miner does Solve the problem ______________

Question No. 6 and 10 carry FIVE marks each **(5x5 =25)**

6. Analyze how decentralization, transparency, security and immutability together strengthen trust in financial transactions.
7. State the features of cryptography.
8. Construct a merkle tree using the hash function 'key mod 5' for the following input 364, 225, 618, 992, 548, 28.
9. In the Bitcoin network, assume the following three users: User A: trader who frequently buys and sells Bitcoin daily; User B: A long-term investor planning to hold Bitcoin for 5 years; User C: A non-technical user who wants convenience for small payments.
   i. Evaluate which wallet type (hot, cold, custodial, non-custodial, hardware, etc.) is most suitable for each user.
   ii. Justify your choice by comparing security, accessibility, and risk factors for each case.
   iii. Suggest a possible hybrid strategy for maximizing both convenience and security.
10. Discuss the types of nodes in bitcoin network.

Question No. 11 and 12 carry TEN marks each **(10x2 =20)**

11. Explain the process of creating a simple blockchain network consisting of 3 nodes, where each node generates 5 transactions. Your answer should include the following:
    a) Define the role of the genesis block and explain how its address (hash) is established.
    b) Describe the structure of a block and how the current block and the next block are linked using block addresses.
    c) List the key inputs contained in each transaction, and explain their significance.
    d) Illustrate how transactions from different nodes are propagated, validated, and added to the blockchain.
    e) Explain how the integrity and immutability of the blockchain are maintained through cryptographic hashing and linking blocks.
12. Determine the points on the curve $y^2 = x^3+ax+b \bmod 11$, for set of all integer and show that point (3,8) falls on the curve.

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End Semester 2025 (Monsoon) CS 425 Blockchain Technology

Date: 15-12-2025 & Time: 10:00 AM to 01:00 PM **Max Marks: 50**

Question No. 1 Carry ONE marks each (Answer ALL) **(1x10=10)**

1. **MCQ**

i) A blockchain is

A) A centralized database that holds a subset of all transactions on all nodes.
B) A client-server database existing on a limited number of nodes at the same time.
C) A distributed database with a record of all transactions on the network.
D) A standalone database with history of all transactions on various nodes.

ii) Advantage of a public blockchain:

A) It does not use disinterested third parties to secure blocks, as all participants have a vested interest.
B) It is more resilient against fraud, because it uses federated nodes to combat fraud.
C) It is open to everyone in the world without permission and licensing requirements.
D) Its networks are built by for-profit companies and the working of the network is guaranteed.

iii) The information is secured in a blockchain

A) By using a closed peer-to-peer (P2P) network, sharing information across platforms
B) By using a distribution of cryptocurrencies over miners through the network
C) By using asymmetric cryptography, consisting of a public and private key
D) By using distributed ledger technology (DLT), which records transactions at the source

iv) The task of miners in blockchain network?

A) Miners act as a single third party to aggregate records and provide trust in the network by the miners' authority.
B) Miners are computers that allow access to the blockchain, ensuring the number of corrupt nodes will stay low.
C) Miners are nodes that compete for a reward by calculating the correct nonce to make a transaction possible.
D) Miners determine the consensus rules that should be followed and interfere when these rules are broken.

v) Define a node

A) The first block of a blockchain
B) A computer on a blockchain
C) A line in a distributed ledger
D) An exchange of cryptocurrencies

vi) A block in a blockchain consists of

A) Hash point, time stamp, transaction data
B) Hash point, IP of owner, transaction data
C) Blockchain name, IP of owner, transaction data
D) Hash point, time of stamp, IP of owner

vii) A blockchain configuration that allows participating members to come from within its organization defines

A) Public blockchain technology.
B) Consortium blockchain technology.
C) Private blockchain technology.
D) Federated blockchain technology.

viii) When a record is submitted in a blockchain, how can you alter it

A) You need the public and private key to modify the record
B) Once submitted, records cannot be altered
C) Only a miner has the right to modify a record in a certain timeframe
D) Only smart contracts have the right to modify the record

ix) **A smart contract is**	x) **Ethereum is**
A) A smart contract is a computer protocol intended to digitally facilitate, verify, or enforce the negotiation or performance of a contract B) Computer code running on top of a blockchain containing a set of rules under which the parties to that smart contract agree C) The two definitions here under are correct D) None of the definitions here under are correct	A) The name of a public blockchain B) The name of a peer to peer network C) A commercial blockchain-based protocol featuring a smart contract manager D) An open-source blockchain-based distributed computing platform featuring smart contract functionality.

Question No. 2 to 5 carry FIVE marks each **(4x5 =20)**

2. Explain step-by-step procedure for doubling a point $P(2P)$ on the ECC curve.
3. Alice wants to buy a token from a decentralized exchange (DEX) at a low price. She submits a transaction with a normal gas fee. At the same time, another user, Bob, notices Alice's transaction in the public mempool and submits a similar transaction but with a higher gas fee. Since miners prioritize transactions with higher gas fees, Bob's transaction is included in the block first. As a result, the token price increases before Alice's transaction is executed, causing her transaction to fail or become more expensive.
   i) What role does gas price play in determining which transaction is included first in the block?
   ii) Why does Alice's transaction fail or delay?
   iii) Explain why Alice's transaction might cost more gas if it is executed after Bob's transaction.
4. Detail the properties of Smart Contract.
5. Design a workflow for a blockchain system that uses SHA-256 to ensure data integrity. Illustrate how hashing is applied at different stages (transaction creation, block formation, and verification).

Question No. 6 to 8 carry TEN marks each (Answer any Two) **(10x2 =20)**

6. A healthcare consortium wants to securely share patient medical records across multiple hospitals while ensuring privacy, auditability, and tamper-proof storage. They are considering implementing a blockchain-based system where each hospital can add patient data, and authorized users (doctors, insurers) can access it based on permissions. Analyze how blockchain properties such as immutability, transparency, decentralization, and consensus mechanisms can address the challenges of sharing medical records.

7. Design a smart contract for a university to automate the certificate issuance process using blockchain technology, focusing on access control and security. The contract should store certificate details, including studentID, courseName, and issueDate. Only an authorized admin should be able to issue (add) new certificates. Anyone should be able to verify whether a certificate exists for a given studentID and courseName. The contract should also prevent issuing duplicate certificates for the same student and course.

8. An agricultural export company wants to increase transparency, reduce fraud, and automate payments in its supply chain using the Ethereum blockchain. Each shipment moves through multiple checkpoints from farmers, warehouses, quality inspectors, transporters, and exporters. Explain how Ethereum's architecture (accounts, transactions, EVM, gas mechanism) supports building a decentralized supply chain system.

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